DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

ENTERPRISE INFORMATION SYSTEMS

Previous Year Questions with Answers: Unit I

Course Title	Enterprise Information Systems	**Course Code**	IM355TBB
Semester	V	**Units Covered**	Unit I
Compiled From	CIE and SEE papers, 2024–2026	**Questions**	Part A: 19 Part B: 13

Notes:

1. Questions are compiled from the CIE and SEE question papers of 2024–2026. The source paper is shown below each question; a question that was asked more than once is listed once with all its sources.
2. Part A contains fill-in-the-blank and short-answer questions. Part B contains descriptive, application and case-based questions (8 to 10 marks).
3. Answers are written strictly from the Unit I textbook material (Analysis and Framework to EIS; Enterprise System Development and Deployment; ERP, CRM, SCM, Decision-Making and Digital Transformation). No outside examples or facts are added.
4. Unit I syllabus: Introduction to Enterprise Systems, Overview of Enterprise Information Systems (EIS), Key components: ERP, CRM and SCM, Role of EIS in business decision-making, Introduction to digital transformation.
5. M: Marks; BT: Bloom's Taxonomy level; CO: Course Outcome.

Sl. No.	Questions	M	BT	CO
	PART – A			
1	List two key features of a Customer Relationship Management (CRM) system. Source: CIE-I, 27/11/2024; SEE Feb/Mar 2025, Q1.1	02	2	1

- **Operational CRM:** automates customer-facing processes through sales force automation, marketing automation and service automation (customer data tracking, lead follow-up, order management and service requests).
- **Analytical CRM:** analyses customer data with data mining, business intelligence and predictive analytics to understand customer behaviour and preferences (segmentation, profitability and lifetime-value analysis).

Other features: collaborative CRM (sharing customer information across departments), strategic CRM (customer retention and loyalty) and campaign management; an integrated CRM gives a 360-degree view of customer data.

2	Mention two types of data commonly handled by EIS. Source: CIE-I, 27/11/2024; SEE Feb/Mar 2025, Q1.2	02	2	1

- **Data held in databases:** records of the enterprise database system, business objects of the ERP system and transaction data from legacy and packaged applications.
- **Unstructured data and text:** XML documents, e-mail and web content.

3	Name two popular ERP software solutions. Source: CIE-I, 27/11/2024	02	1	2

SAP (SAP-AG, the pioneer and leading vendor of ERP/EIS systems) and **Oracle**. (Other vendors named in the textbook: PeopleSoft, BaaN, JD Edwards and Microsoft Dynamics.)

4	What is the primary function of Supply Chain Management (SCM) within EIS? Source: CIE-I, 27/11/2024	02	1	1

SCM integrates and coordinates all activities related to the **flow of goods, services, information and finances from raw-material suppliers to the final consumer**. Its main objective is to deliver the right product, in the right quantity, at the right time and at the right cost while maintaining efficiency and customer satisfaction.

5	What are the core modules of an ERP system? Source: CIE-I, 27/11/2024	02	2	2

The core modules of an ERP system are:

- Financial management and Human resource management (HRM)
- Supply chain management (procurement, logistics, production planning)
- Manufacturing and production management; Inventory and warehouse management
- Customer relationship management (CRM); Project management
- Business intelligence (BI) and reporting

In SAP terms: FI, CO, AM, PS, HR, PM, MM, QM, PP, SD and CA.

6	_____ characteristic is typically associated with an Enterprise Information System (EIS). Source: SEE Apr/May 2024 (21IM55B2), Q1.1	01	1	1

Answer: Integrated (an EIS is an integrated system shared by all users)

7	_____ is the main benefit of EIS from a technical standpoint. Source: SEE Apr/May 2024 (21IM55B2), Q1.2	01	2	2

Answer: Integration (data integration): data are entered once at the most accurate source and shared by all users, which eliminates sub-organizational silos

8	List two fundamental ways an Enterprise Information System contributes to improving an organization's operational efficiency. Source: SEE Feb/Mar 2026, Q1.1	02	1	1

- **Automation of routine processing:** EIS automates routine data processing, which reduces redundancy and errors and streamlines workflows, so operational decisions are made faster (example: automatic purchase-order generation when stock falls below a threshold).
- **Real-time information and monitoring:** EIS collects and organizes enterprise-wide data in real time and lets managers track KPIs and respond quickly to deviations (example: real-time alerts about shipment delays).

9	Enterprise Systems are noted for providing a tightly integrated solution to an organization's information needs. State two functions that were historically stand-alone applications. Source: CIE-I, 10/11/2025	02	2	2

Inventory control, sales, receivables and financial accounting were stand-alone applications that are integrated in an EIS. (The old isolated systems in finance, human resources, manufacturing and warehousing are likewise replaced by one unified program.) Any two of these.

10	The primary mechanism by which Enterprise Information Technology provides value to organizations is through appropriate _____ improvement. Source: CIE-I, 10/11/2025	01	1	1

Answer: business process (improvement): efficient, integrated information systems lead to more efficient business processes, which make the company more competitive

11	What is the role of EIS in business decision-making? State one key function. Source: CIE-I, 10/11/2025 (Q4 and Q6)	02	1	1

EIS integrates the processes and data of finance, production, marketing and HR into one framework and gives managers at the strategic, tactical and operational levels timely and accurate information for informed decisions.

One key function: enhancing information availability. EIS collects, stores and organizes enterprise-wide data in real time, so managers have accurate, up-to-date information when making critical decisions (example: an ERP dashboard showing real-time sales, inventory and production data).

12	State how data from an Enterprise Information System (EIS) supports strategic-level decision-making. Source: SEE Feb/Mar 2026, Q1.6	02	2	2

Through the integrated ERP, CRM and SCM modules, EIS provides insight into market trends, customer behaviour and supply-chain performance. Top management uses forecasting tools, trend analysis and dashboards for long-term strategic planning (example: forecasting demand patterns and adjusting capacity or investment decisions accordingly).

13	How does the implementation of an integrated EIS/ERP system (like the one used in the Sanyo case) achieve “process quality improvement”? Source: SEE Feb/Mar 2026, Q1.8	02	1	2

An integrated EIS/ERP improves process quality because it:

- implements better ways of accomplishing tasks (best practices), which gives better operational quality and greater productivity;
- models the business processes electronically, so an order can be tracked from the order desk through production planning, purchasing, production, warehousing, shipping and accounts receivable (order to payment);
- enters data once at the most accurate source and shares it, so data are more complete and accurate.

Note: The Sanyo case is not described in the textbook; the answer applies the textbook's general reasoning on EIS and process quality.

14	A textile company is implementing a new EIS. Its goal is to maximize the utilization of collective employee knowledge for project teaming and innovation. Which core EIS component is primarily responsible for managing and exploiting this intellectual capital? Justify your choice. Source: CIE-I, 10/11/2025	02	3	3

Knowledge management (knowledge work / portal-based knowledge system). Employee expertise is one of an organization's most valuable assets but one of the most difficult to exploit. In the Ericsson case, the knowledge portal profiled users automatically and helped create strong project teams by identifying employees with relevant internal competence, so the intellectual capital was exploited and productivity rose. Knowledge work systems likewise support knowledge workers in the creation and integration of new knowledge in the organization.

15	Define the term "Digital Transformation" (DT). Source: SEE Feb/Mar 2026, Q1.2	02	1	1

Digital transformation (DT) is the strategic adoption of digital technologies (cloud computing, AI, IoT, data analytics) to radically improve business processes, culture and customer experiences. It goes beyond mere digitization: it redefines how an organization operates and creates value in a hyper-connected, data-driven world.

16	The evolution of planning systems progressed from MRP-I to _____ and then to ERP. Source: CIE-I, 10/11/2025	01	1	1

Answer: MRP-II (Manufacturing Resource Planning): MRP-I, then MRP-II, then ERP

17	What is the primary purpose of Material Requirements Planning (MRP)? Source: SEE Apr/May 2024 (21IM55B2), Q1.3	01	1	2

MRP is a time-phased order-release system. Its purpose is to plan production and raw-material requirements by working backwards from the sales forecast, so that work orders and purchase orders are scheduled and released and sub-assemblies and components arrive at the assembly station just as they are required. It tackles the problem of dependent demand and gives the benefits of lower inventories, improved customer service and greater efficiency.

18	What does MRP-3 (Money Resource Planning) primarily focus on? Source: SEE Apr/May 2024 (21IM55B2), Q1.16	01	2	1

MRP-III (Money Resource Planning, also called ERP-II) emphasizes the **planning of capital (money) in an optimal manner**, including managing the situation when surplus money arises.

19	List a common challenge faced during ERP implementation related to software customization. Source: SEE Apr/May 2024 (21IM55B2), Q1.4	02	2	1

Organizations must decide the degree of customization. The best fit needs intensive reengineering and software development by internal staff, which is unrealistically slow and expensive, whereas adopting the vendor's software as it is faster and less troublesome. In practice almost every organization must modify the vendor software to some degree. Customization adds time and cost to implementation and dilutes the integration benefits: the more an ERP is customized, the less it communicates seamlessly within the system and with supplier and customer systems.

Sl. No.	Questions	M	BT	CO
	PART – B			
1	Analyse the benefits and challenges of implementing an EIS in a manufacturing organization. Support your answer with examples. Source: CIE-I, 27/11/2024; SEE Feb/Mar 2025, Q2a	10	2	1

An EIS is an enterprise-wide integrated information system that covers all functional areas (planning, manufacturing, sales and marketing), performs the core corporate activities and uses integrated database systems. Its pros and cons are summarized in the table.

Factor	Benefit (pro)	Challenge (con)
System integration	Improved understanding across users	Less flexibility
Data integration	Greater accuracy	Harder to make corrections; imposition of how people do their work
Best practices	More efficient methods	Less freedom and creativity; changing needs
Cost of computing	More efficient system planned	Under-budgeted training expense; hidden costs of implementation

Benefits in a manufacturing organization

(1) **Integrated system:** one system shared by all users replaces a diverse set of applications that rarely communicate with each other; the main benefit is the elimination of sub-organizational silos that serve only their own problems.
(2) **Data integration and accuracy:** data are entered once at the most accurate source and shared by all users, so data are more complete and accurate.
(3) **Best practices:** better ways of doing things give better operational quality and greater productivity.
(4) **Planning and inventory control:** MRP/MRP-II functions give better inventory and raw-material control, reduced inventories, shorter lead times for materials, improved customer service and less clerical support.
(5) **Order tracking:** a sales order can be tracked from the order desk through production planning, purchasing, production, warehousing, shipping and accounts receivable, and its effect on purchasing and production capacity can be seen.
(6) **Strategic gains:** improved decision-making, support for business growth, external linkages with suppliers and customers, and better customer service; lower IT cost and greater flexibility in the long run.

Example from the textbook case: the State of South Carolina's pilot eliminated a purchasing backlog of about one week and documented process savings of about $600,000 a year.

Challenges

(1) **Less flexibility:** the EIS imposes one procedure on the whole organization and everyone must conform to the new system.
(2) **Harder corrections:** controls are needed so that changes do not introduce new errors.
(3) **Best practices need effort:** identifying the best way for each function takes a great deal of effort and often means significant change in how people work; what one considers best is not best for all.
(4) **Cost:** full ERP systems cost from about $5 million for very small versions to over $100 million for very large ones; training is typically under-budgeted and the hidden cost of lost productivity is usually ignored.
(5) **Expected savings are often not realized,** because all user needs and later changes in the business environment were not anticipated.
(6) **Implementation problems:** loss of control through inadequate project definition and planning; people who followed the same practices for years are slow to accept change; customization of vendor software.

Conclusion: the benefits of integration are usually much greater than the cost of conformity.

2	How can information technology support a company's business processes and decision making and give it competitive advantage? Give examples to illustrate your answer. Source: CIE-I, 27/11/2024; SEE Feb/Mar 2025, Q2b	10	3	3

Information technology (computing technology such as networks, hardware, software and the Internet) is the means to deal with and control the information resource. The right information delivered to the right person, in the right fashion and at the right time makes organizational effectiveness and competence more certain.

1. Supporting business processes

- An EIS models the core and supportive business processes of the company electronically and gives a global view of the organization, with less duplication and fewer errors.
- An EIS can track a sales order from the order desk to production planning, purchasing, production, warehousing, shipping and accounts receivable (order to payment) and show its impact on other departments.
- Electronic data interchange (EDI) lets companies handle purchasing electronically, avoiding the cost and delay of paper purchase-order and invoice systems.
- **Examples of EIS in use:** airline reservations, bank operations (deposit, transfer, withdrawal) with payment gateways, logistics management applications and IRCTC train reservation.

2. Supporting decision-making

- The types of information systems support the levels of the organization: transaction processing systems (routine items), management information systems (focused information flow for managing the business), decision-support systems (models for exceptional, non-recurring, moderately unstructured decisions) and executive support systems (strategic, non-customized decisions through advanced graphics and communications).
- EIS makes real-time enterprise data available, supports strategic planning from market, customer and supply-chain insights, and uses BI dashboards to highlight underperforming products or regions and to monitor KPIs.

3. Competitive advantage

- Companies must give customers goods and services faster and cheaper than the competition; efficient, integrated information systems give more efficient business processes, and companies with efficient processes are more competitive.
- If competitors adopt ERP and gain cost efficiencies and serve customers better, an organization is left with a declining clientele; ERP/EIS also improves interaction with suppliers and customers, which is a way to gain strategic advantage.
- **Example (Ericsson):** a single company-wide portal gave speedier access to competitive information and to internal expertise; productivity and quality of work rose and responsiveness in the marketplace improved.
- **Example (Amazon):** its use of AI and data analytics in supply chain and customer service has set industry benchmarks for digital excellence.

3	Define and differentiate the core components of EIS. Briefly define ERP, CRM and SCM, and explain the unique focus of each component within the overall enterprise system framework. Source: CIE-I, 10/11/2025	10	2	2

ERP, CRM and SCM are the three enterprise-class applications that make up the EIS. An EIS includes the internal back-office operations and the external front office, such as the financial system, human resources, inventory management, shipping, customer order processing and warehouse. The back office looks after raw material, financial issues, logistics and other internal matters; the front office interacts directly with the customer (CRM is a front-office package).

Basis	ERP	CRM	SCM
Definition	An integrated software system that unifies business functions across departments into a single database and user interface, so that information flows seamlessly between business processes.	A strategic approach that integrates technology, processes and people to manage an organization's interactions with current and potential customers.	Integration and coordination of all activities related to the flow of goods, services, information and finances from raw-material suppliers to the final consumer.

Unique focus	Integration of the internal functions of the enterprise (back office) on a common database.	The customer: satisfaction, retention and profitability (front office).	The flow from suppliers to customers: right product, quantity, time and cost.
Components	Financial, HRM, SCM, manufacturing and production, CRM, inventory and warehouse, project management, BI and reporting.	Operational, analytical, collaborative, strategic and campaign management CRM.	Planning, sourcing, manufacturing, distribution, return management, integration and coordination.
Benefits	Data accuracy, productivity, cost and time savings, audit readiness, decision support.	360-degree customer view, satisfaction and loyalty, faster response, lower service cost.	Lower operating cost, better supplier relations, flexibility, competitive advantage.
Example	Automated generation of balance sheets; salary records updated from attendance.	Sales tracking and follow-up reminders (Salesforce).	Real-time tracking of raw materials from vendor to warehouse.

Framework: the three components are linked through a common database so that sales, production, customer and supplier information is shared; ERP-based demand forecasting drives SCM planning, and ERP-CRM integration gives sales teams customer purchase histories.

4	Provide a summary of the features of various Enterprise Resource Planning (ERP) modules. Source: SEE Apr/May 2024 (21IM55B2), Q2b	08	2	1

ERP software is sold in modules that share one database; most vendors let a firm install some modules without buying the whole package. The SAP modules (with their Oracle equivalents) are:

Code	Module	Features	Oracle equivalent
FI	Financial accounting	General ledger, account transactions, generates financial statements	Payables, receivables, general ledger, cash management
CO	Controlling	Internal management; cost analysis by cost centre	Treasury
AM	Asset management	Fixed-asset purchase and depreciation	Spares and enterprise asset management
PS	Project system	Manage projects	Projects
HR	Human resources	Recruiting, hiring, training, payroll, benefits	Advanced benefits, time and labour, human resources
PM	Plant maintenance	Preventive maintenance and resource management	Assets
MM	Materials management	Purchasing and raw-material inventory, work in process, finished goods	Bills of material, flow manufacturing, master scheduling/MRP
QM	Quality management	Product inspections, material certifications, quality control	Quality
PP	Production planning	Production planning and scheduling, actual production	Supplier scheduling
SD	Sales and distribution	Sales orders, scheduled deliveries, customer information	Sales, order capture, TeleSales
CA	Cross application	Workflow, business information warehouse, e-mail, industry solutions, CRM, SRM	CRM gateway, customer/sales/marketing intelligence, supply chain trading

SAP also groups its offerings as solutions: financials, human resources, CRM, supplier relationship management, product lifecycle management, supply chain management and business intelligence.

Functional view of the ERP components

Component	Key functions
Financial management	General ledger, payables and receivables, budgeting and forecasting, asset management and financial reporting
Human resource management	Recruitment, onboarding, payroll, attendance, leave and performance tracking, training
Supply chain management	Procurement and inventory management, supplier management and logistics, demand forecasting, order tracking
Manufacturing and production	Bill of materials, production scheduling, shop-floor control, maintenance, quality assurance, cost estimation
Customer relationship management	Lead management and sales tracking, customer service and feedback, marketing campaigns
Inventory and warehouse	Stock tracking and control, barcode-based material movement, warehouse space optimization
Project management	Project planning and budgeting, resource allocation, progress and milestone tracking
BI and reporting	Real-time analytics, customizable dashboards and KPIs, predictive analysis and forecasting

5	Sketch the demand-supply chain service provider model. Explain the concepts of Enterprise Information Systems (EIS). Source: SEE Apr/May 2024 (21IM55B2), Q2a	08	2	2

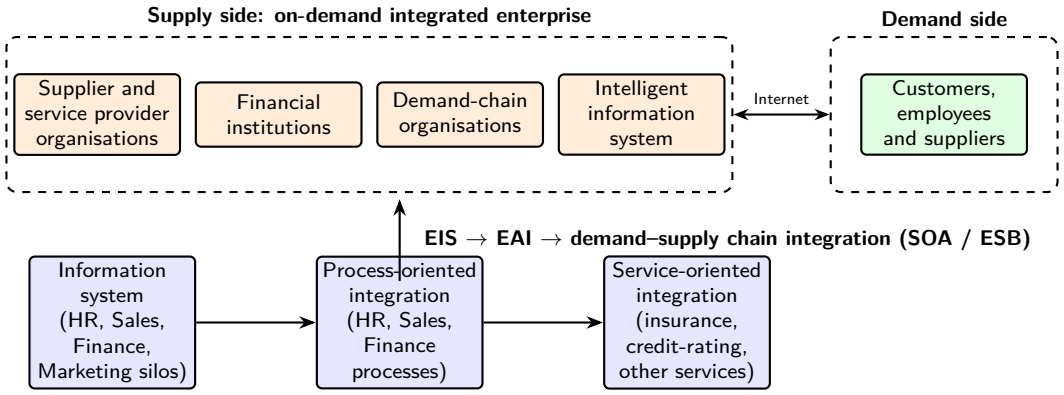


Figure 1: Enterprise demand–supply chain (service provider) model

The model: on the supply side the on-demand integrated enterprise links supplier and service-provider organizations, financial institutions, demand-chain organizations and an intelligent information system; the demand side consists of customers, employees and suppliers. The information systems evolve from separate functional systems (HR, sales, finance, marketing) through process-oriented integration to service-oriented integration (SOA/ESB), in which services such as insurance and credit rating are linked, giving demand-supply chain integration.

Concepts of EIS

- **Integrated platform:** an EIS is a technology platform that enables organizations to integrate and coordinate their business processes; it provides a single system, central to the organization, that shares information across all functional levels and management hierarchies and removes information fragmentation by creating a standard data structure.
- **Enterprise-wide, not only manufacturing:** it covers all functional areas and core corporate activities (planning, manufacturing, sales and marketing), integrates legacy systems into one coordinated system and uses integrated database systems.
- **Reengineering of the business:** an EIS reengineers the manner in which an organization accomplishes business tasks and transforms the organization from a functional into a process-centric entity.
- **Planning of the 5 Ms:** it plans the enterprise's Man, Money, Material, Method and Machinery to their best synergistic values.

- **Right information at the right time:** integrated, uniform, relevant and up-to-date information gives the right person authority to make decisions at the right time.
- **Back office and front office:** it includes the internal back-office operations and the external front office such as customer order processing; it is one of three enterprise-class applications with CRM and SCM.
- **Common features:** best business practices, connectivity beyond the enterprise, comprehensiveness, flexibility and a modular, open architecture.

6	Distinguish between MRP-I and MRP-II. Explain the primary functions of each planning system and illustrate how the evolution from MRP-I to MRP-II broadened the scope of enterprise planning. Source: CIE-I, 10/11/2025	10	2	2

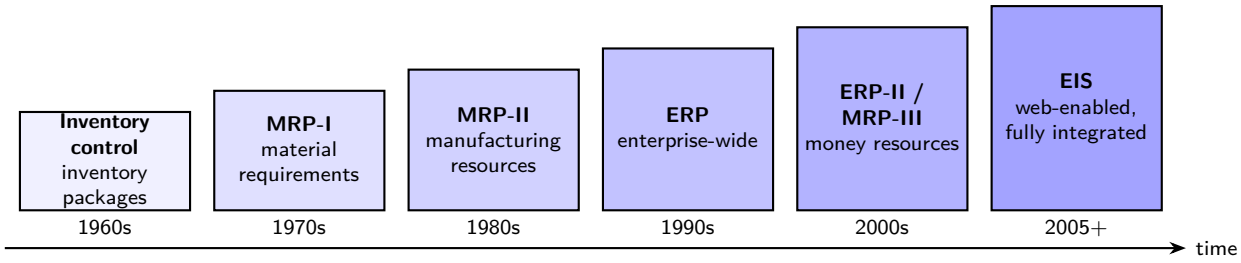


Figure 2: Evolution of planning systems: inventory control, MRP, MRP-II, ERP and EIS

MRP-I (Material Requirements Planning)

MRP is a planning instrument geared exclusively to assembly operations and was evolved to tackle the problem of dependent demand. It begins with three documents: a forecast for end items (master production schedule), the bill of materials (BOM) and the inventory records. It tells when to place the order, how much to order, whom to order from and when the items must be on hand.

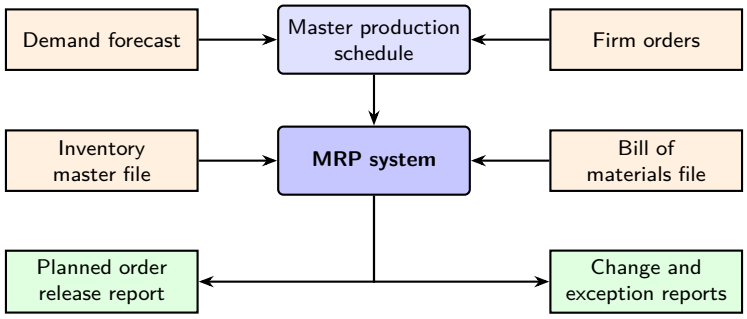


Figure 3: MRP documents: inputs and outputs of an MRP system

Features: focus on time delivery; in-house and vendor lead times are incorporated; time-phased calculation of procurement quantities and dates; reduced inventory levels. **Outputs:** planned orders, order release notices, action notices to expedite or cancel orders, priority reports, inventory status and performance reports. **Benefits:** reduction of inventories, improved customer service and enhanced efficiency.

MRP-II (Manufacturing Resource Planning)

MRP-II is a method to plan all the resources of a manufacturer. A variety of functions are tied into it: order processing, business planning, sales and operations planning, production plans, master production scheduling, capacity requirements planning and capacity planning. It is integrated with accounting and finance to produce reports such as business plans, shipping budgets, inventory projections and purchase plans. Its major purpose is to integrate the primary functions (production, marketing and finance) and the other functions (personnel, engineering and purchasing) into the planning process to improve the efficiency of the manufacturing enterprise.

Basis	MRP-I	MRP-II
Full form	Material Requirements Planning	Manufacturing Resource Planning
Period	1960-70 (MRP systems emerged in the 1970s)	1980 (MRP extended to the shop floor and distribution management in the 1980s; JIT methodology of the late 1980s)

Purpose	Tap inventory, i.e. raw-material planning; solves dependent demand	Plan all resources of a manufacturer; looks after production-related activities, shop floor and distribution management
Scope	Materials and procurement (with capacity planning and shop floor in closed-loop MRP)	Closed-loop MRP integrated with other functions such as finance and marketing
Information	Forecast, bill of materials and inventory records	Operational information stored centrally and shared with other functional areas
Outputs	Planned orders, release and action notices, priority reports	Business plans, shipping budgets, inventory projections, purchase plans

How the evolution broadened enterprise planning

Manufacturing software developed from simple inventory tracking systems (1960s) to MRP (1970s), which was extended to the shop floor and distribution management (1980s MRP-II) and then to finance, marketing and other functions. This led to ERP (1990s), covering engineering, finance, human resources and project management as a whole, then to MRP-III (money resource planning) in the 2000s, and to the web-enabled, fully integrated EIS (2005).

7	Evaluate the role of EIS in Customer Relationship Management (CRM) with specific examples of how companies have leveraged EIS for CRM. Source: CIE-I, 27/11/2024	10	2	3

CRM is a strategic approach that integrates technology, processes and people to manage interactions with current and potential customers, to improve customer satisfaction, retention and profitability. In an EIS, CRM is the front-office application that works with the integrated back-office data of ERP and SCM.

Role of EIS in CRM

(1) **Integrated customer data:** a common database gives a 360-degree view of the customer; ERP-CRM integration gives sales teams the customer purchase histories.
(2) **Operational CRM:** automates sales force, marketing and service processes (lead generation and follow-up, order management and service requests).
(3) **Analytical CRM:** data mining, BI and predictive analytics on the collected data reveal customer behaviour, segments and profitability (example: analysing buying patterns to design loyalty programmes).
(4) **Collaborative CRM:** shares customer insights across sales, marketing and service and across channels (example: a support team accessing customer order history from the sales database).
(5) **Strategic and campaign CRM:** long-term retention and loyalty plans; planning, running and measuring campaigns (example: customized e-mail offers to a customer segment).

How companies have leveraged EIS for CRM

- **Salesforce:** automates sales tracking and follow-up reminders.
- **Oracle and SAP:** Oracle purchased Siebel Systems and made its CRM product a module of its system; SAP responded by developing its own CRM, so what had been an external add-on became an internal module.
- **Amazon:** uses AI and data analytics in its supply chain and customer service; integrating AI into CRM allows predictive analysis of customer behaviour.

Benefits: 360-degree customer view, enhanced customer satisfaction and loyalty, improved sales and marketing effectiveness, reduced response time and service cost, and data-driven strategic decisions.

8	Describe the hierarchical role of an Enterprise Information System (EIS) in supporting decision-making across the three organizational management levels: strategic, tactical and operational. Source: SEE Feb/Mar 2026, Q2b and Q6a (same question, reworded)	08	1	2

An EIS integrates processes and data of all departments into one framework and supports managers at all levels (strategic, tactical and operational) with timely and accurate information.

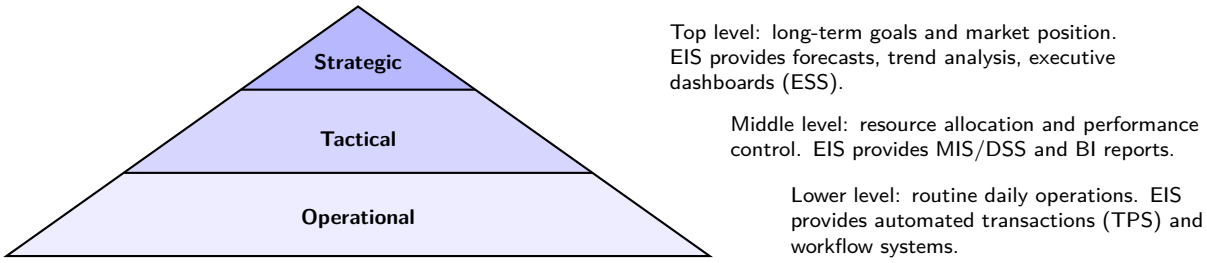


Figure 4: Levels of decision support provided by an EIS

Level	Type of decision	EIS contribution	Systems and example
Strategic (top level)	Long-term goals, market positioning	Forecasting tools, trend analysis, dashboards; insight into market trends, customer behaviour and supply-chain performance	Executive support systems; executives forecast demand patterns and adjust capacity or investment decisions
Tactical (middle level)	Resource allocation, performance control	Decision-support systems and BI reports; real-time monitoring of KPIs	MIS and DSS; BI dashboards highlight underperforming products or regions to guide marketing
Operational (lower level)	Routine daily operations	Automated transactions and workflow systems, which reduce redundancy and errors	Transaction processing systems; automatic purchase order when stock falls below a threshold

Because departments share the same databases, information flows consistently and transparently and decisions are aligned across functional areas. The integrated modules (ERP for cross-functional processes, CRM for customer insight, SCM for inventory and logistics) together give comprehensive, data-backed decision-making with improved accuracy and speed.

9	Explain the scope of Digital Transformation in the context of EIS. Describe what the term means for a modern enterprise and how the adoption of an integrated EIS serves as the foundational enabler for achieving digital transformation goals. Source: CIE-I, 10/11/2025	10	2	1

Digital transformation (DT) is the strategic integration of digital technologies into all areas of an organization to fundamentally change how it operates and delivers value to customers. It goes beyond mere digitization: it redefines how an organization creates value in a hyper-connected, data-driven world, using cloud computing, AI, IoT and data analytics.

Scope of DT for a modern enterprise (core dimensions)

Dimension	Description
Technology	Integration of cloud computing, AI, IoT, big data and mobile platforms into enterprise systems
Process	Reengineering workflows using automation, RPA and agile methodologies to improve speed and efficiency
People	Empowering employees with digital tools, fostering a culture of innovation and upskilling for digital fluency
Strategy	Aligning digital initiatives with business goals to drive competitive advantage and customer-centricity

Integrated EIS as the foundational enabler

Enterprise information systems are the backbone of digital transformation. They enable:

(1) **Real-time data access** for informed decision-making.
(2) **Cross-functional integration** across the ERP, CRM and SCM modules on a common database.
(3) **Scalable infrastructure** through cloud-based platforms.
(4) **Enhanced customer engagement** through personalized services and omnichannel experiences (example: AI integrated into CRM for predictive customer-behaviour analysis; IoT-enabled SCM optimizing inventory and logistics in real time).

Business impact: operational efficiency, agility and innovation, customer-centricity and a data-driven culture. **Challenges:** legacy-system integration, cybersecurity risks, change management and employee resistance, and data privacy and compliance.

10	Explain the concept of Digital Transformation (DT). Describe the crucial role of two key technologies, Cloud Computing and Artificial Intelligence (AI), in driving this transformation within a modern enterprise. Source: SEE Feb/Mar 2026, Q2a	08	1	2

Digital transformation (DT) is the strategic adoption of digital technologies to radically improve business processes, culture and customer experiences, redefining how an organization creates value in a hyper-connected, data-driven world. It modernizes processes and systems, uses data for decisions and adopts digital platforms like ERP, CRM and SCM to unify operations.

Role of cloud computing

- Enables flexible, scalable and cost-effective deployment of enterprise applications.
- Provides the scalable infrastructure on which EIS runs through cloud-based platforms.
- Supports global collaboration and scalability (a supporting technology of modern SCM).

Role of artificial intelligence

- Powers intelligent automation, predictive analytics and natural language processing.
- Integrating AI into CRM allows predictive analysis of customer behaviour; in SCM, AI and analytics predict demand and optimize routes.
- **Example:** Amazon's use of AI and data analytics in its supply chain and customer service has set industry benchmarks for digital excellence.

Together: cloud computing supplies the scalable platform for the enterprise systems and AI turns their data into automation and predictive insight, giving operational efficiency, agility and innovation, and customer-centric, data-driven decisions.

11	You are advising a high-growth tech startup. The founders are considering which EIS component to prioritize first: CRM for managing investor relations and early customer feedback, or a modular ERP for integrating the finance and HR functions. Argue for the strategic component that should be prioritized to support the startup's growth and eventual digital transformation. Source: CIE-I, 10/11/2025	10	3	3

Recommendation: prioritize the modular ERP (finance and HR first); add CRM afterwards as a module of the same system.

Arguments

(1) **ERP is the foundation:** ERP integrates the old isolated systems of finance, human resources, manufacturing and warehousing into a single unified program on one common database, and it is the building block on which other elements that give competitive advantage are added.
(2) **It is modular:** most vendors' ERP software is flexible enough to install some modules without buying the whole package, and many companies immediately install an ERP finance or HR module and leave the rest for later.
(3) **Cost and risk:** full ERP is very expensive; the fewer modules, the lower the cost. Implementation projects rarely go as planned or stay within budget, so trying particular modules first and seeing how the organization digests them reduces risk (phased rather than big-bang implementation).

(4) **Finance is easiest to implement:** financial and accounting modules are typically installed first because they involve the most structured application; they are also the most widely used modules. Modules supporting less structured activities such as sales and marketing, where CRM lies, have encountered notable implementation difficulty.
(5) **Digital transformation:** an integrated EIS gives real-time data, cross-functional integration and a scalable base on which the later CRM, SCM and analytics are built.

Place of CRM

CRM (investor relations, customer feedback, 360-degree customer view) is valuable, but it is a front-office application that works best when integrated with the back-office data. Vendors have made CRM an internal module of their ERP suites, so it can follow the finance and HR modules without a separate system; ERP-CRM integration then gives sales teams the customer purchase histories.

12	Highlight some of the issues you have faced in using SAP implemented in your institution. How did you resolve those issues? Source: CIE-I, 27/11/2024	10	3	2

The textbook does not describe any particular institution's SAP system, and SAP is its example of a leading ERP/EIS vendor. The answer therefore gives the typical issues the textbook identifies in using and implementing an ERP/EIS such as SAP, and how they are resolved.

Issue	Reason (textbook)	How it is resolved
Rigid procedures; users must conform	The system imposes one procedure on the whole organization; less freedom and creativity	Accept that the benefits of integration are usually greater than the cost of conformity; adopt the best practices built into the package
Corrections are difficult	Controls are needed so that changes do not introduce new errors	Enter data once at the most accurate source and follow controlled correction procedures
Inadequate training; lost productivity	Training is under-budgeted and hidden costs arise while staff cope with a complex new system	Budget for training; use consultants to bridge the gap between vendor and user by giving training
Staff slow to accept change	People following the same practices for years resist change; the cultural aspect is the highest hurdle	Change management with accurate and consistent communication, training and risk management
Customization of the software	Almost every organization must modify the vendor software; it adds time and cost	Decide the degree of customization carefully; use middleware for add-on software; adopt vendor software directly where possible
Loss of control, delays and cost overruns	Inadequate project definition, planning and implementation	Clear project definition and planning; top management involvement; a decision based on business needs, not technology
Heavy cost and complexity	Full systems are very expensive and projects rarely go as planned or stay within budget	Implement module by module (phased), starting with financial and accounting modules

13	XYZ Manufacturing Company has been facing inefficiencies in its operations due to a lack of integration between its departments. The procurement, inventory, production and sales teams operate on standalone systems, leading to delays, data inaccuracies and increased costs. The company implemented an ERP system, but after six months it is still struggling with system adoption and is not seeing the expected improvements. Analyze the challenges XYZ Manufacturing is facing in implementing the ERP system. Suggest strategies to overcome these challenges and ensure successful adoption. Source: CIE-I, 27/11/2024	10	4	4

Challenges faced by XYZ Manufacturing

(1) **ERP seen as a magic solution:** the company expects the ERP itself to bring competitiveness, whereas the tool only helps if the company uses it well; misled implementation is a main cause of failure.
(2) **Poor project definition and planning:** inadequate project definition, planning and implementation lead to loss of control.
(3) **Resistance to change:** people used to separate procurement, inventory, production and sales systems are slow to accept change; the cultural aspect of process change is the highest hurdle.
(4) **Inadequate training:** training is typically under-budgeted, and lost productivity while employees cope with the complex new system is a hidden cost.
(5) **Conformity:** the ERP imposes one procedure for the whole organization; the organization's processes must be modified to conform to it.
(6) **Customization and fit:** heavy modification of the ERP adds time and cost and weakens integration.
(7) **All-at-once implementation:** implementing the whole system at one time is difficult; modules for less structured activities such as sales and marketing are harder to implement.

Strategies to overcome them and ensure adoption

(1) **Top-management responsibility and clear project definition:** treat the ERP as a business decision, not a technology decision; define and plan the project properly.
(2) **Change management:** accurate and consistent communication, training and risk management.
(3) **Adequate training budget** and consultants to bridge the gap between vendor and users.
(4) **Phased implementation:** start with financial and accounting and other structured modules, and see how the organization digests them before taking on more (test-the-waters approach).
(5) **Limit customization:** adopt the ERP's best practices and modify the organization's processes to fit; use middleware for add-ons.
(6) **Pilot first and measure benefits:** a pilot gives lessons for roll-out, and a business case and ROI baseline give a yardstick for success (as in the State of South Carolina project).

With one integrated database shared by procurement, inventory, production and sales, data are entered once and shared, which removes the delays and data inaccuracies caused by standalone systems.

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